# Intro to Machine Learning • BA4

## 1 Core Concepts

### Empirical Risk

**Empirical Risk**

$$R(w) = \frac{1}{N}\sum_{i=1}^{N} \ell(\hat{y}_i, y_i)$$

Training $\Leftrightarrow$ $\min_w R(w)$

### Gradient Descent

**Gradient Descent Update**

$$w_k \leftarrow w_{k-1} - \eta \nabla_w R(w_{k-1})$$

$\eta$ = learning rate. Converges when $\nabla_w R(w) = 0$.

**Convergence Criteria**

- $|R(w_k) - R(w_{k-1})| < \delta_R$
- $\|w_k - w_{k-1}\| < \delta_w$
- or fixed number of iterations

### Useful Identities

- $\nabla_b(a^T b) = a$
- $\|a\|^2 = a^T a$
- $\nabla_a \|a\|^2 = 2a$

## 2 Linear Regression

### Model & Training

**Prediction**

$$\hat{y}_i = w^T x_i$$

**MSE Risk**

$$R(w) = \frac{1}{N}\sum_{i=1}^{N}(w^T x_i - y_i)^2$$

**Closed-Form Solution**

$$w^\star = (X^T X)^{-1} X^T y$$

If $X^T X$ not invertible: $w^\star = X^\dagger y$.
**Multi-output:** $W^\star = X^\dagger Y$

### Evaluation Metrics

- **MSE:** $\frac{1}{N}\sum(y_i - \hat{y}_i)^2$
- **RMSE:** $\frac{1}{N}\sum\sqrt{(y_i - \hat{y}_i)^2}$
- **MAE:** $\frac{1}{N}\sum|\hat{y}_i - y_i|$
- **MAPE:** $\frac{1}{N}\sum\left|\frac{\hat{y}_i - y_i}{y_i}\right|$

## 3 Logistic Regression (Binary)

### Model

**Sigmoid**

$$\sigma(z) = \frac{1}{1+e^{-z}}, \qquad \hat{y}_i = \sigma(w^T x_i)$$

### Training

**Cross-Entropy Loss**

$$R(w) = -\sum_{i=1}^{N}\Big[y_i \ln \hat{y}_i + (1-y_i)\ln(1-\hat{y}_i)\Big]$$

**Gradient**

$$\nabla_w R(w) = \sum_{i=1}^{N}(\hat{y}_i - y_i)\, x_i$$

No closed form $\Rightarrow$ use **gradient descent**.

### Prediction

$$\hat{y} = \begin{cases} 1 & \text{if } \sigma(w^{\star T} x) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

## 4 Classification Evaluation

### Confusion Matrix

| | True Pos | True Neg |
|---|---|---|
| **Pred Pos** | TP | FP |
| **Pred Neg** | FN | TN |

$P = TP + FN, \quad N = FP + TN$

### Metrics

- **Accuracy:** $\frac{TP+TN}{TP+TN+FP+FN}$
- **Precision:** $\frac{TP}{TP+FP}$
- **Recall:** $\frac{TP}{TP+FN}$
- **FPR:** $\frac{FP}{FP+TN}$
- **F1:** $2 \cdot \frac{\text{Precision}\cdot\text{Recall}}{\text{Precision}+\text{Recall}}$

**Multi-class**

Use **confusion matrix** (rows = predicted, cols = actual). Accuracy = sum of diagonal / total. Precision/Recall/F1: compute per class then average.

## 5 Multi-Class Classification

### One-Hot Encoding

$y_i \in \{0,1\}^C$: one entry is 1, rest are 0.

### Softmax (Logistic)

**Softmax**

$$\hat{y}^{(k)}(x) = \frac{\exp(w^{(k)T} x)}{\sum_{j=1}^{C} \exp(w^{(j)T} x)}$$

**Multi-Class Cross-Entropy**

$$R(W) = -\sum_{i=1}^{N}\sum_{k=1}^{C} y_i^{(k)} \ln \hat{y}_i^{(k)}$$

**Gradient**

$$\nabla_W R(W) = \sum_{i=1}^{N} x_i\,(\hat{y}_i - y_i)^T$$

**Prediction:** $k = \arg\max_j \ \hat{y}^{(j)}$

## 6 k-Nearest Neighbors

### Distances

- **Euclidean:** $\sqrt{\sum_k (x_i^{(k)} - x_j^{(k)})^2}$
- **Chi-square:** $\sqrt{\sum_k \frac{(x_i^{(k)} - x_j^{(k)})^2}{x_i^{(k)} + x_j^{(k)}}}$

### Classification

Find $k$ nearest neighbors $\Rightarrow$ majority vote.

### Regression

Find $k$ nearest $\Rightarrow$ (weighted) average:

$$\hat{y} = \frac{\sum_i w_i y_i}{\sum_i w_i}, \quad w_i = \frac{1}{d_i}$$

## 7 Feature Expansion & Regularization

### Polynomial Feature Expansion

**Expansion**

$\phi : \mathbb{R}^D \to \mathbb{R}^F$ maps $x$ to higher-dim features.
$\hat{y}_i = w^T \phi(x_i)$

**Closed-Form with Expansion**

$$w^\star = (\Phi^T \Phi)^{-1} \Phi^T y$$

**Cover's Theorem**

A complex classification problem is more likely linearly separable in high-dimensional space (if not densely populated).

### Overfitting & Underfitting

- **Overfitting:** low train error, high test error
- **Underfitting:** high error on both

### Cross-Validation

- **Val. set:** 80/20 split — cheap, wastes data
- **LOOCV:** train on $N-1$, test on 1, repeat — exact, expensive
- **K-fold:** split into $k$ parts, rotate — best compromise

### L2 Regularization (Ridge)

**Regularized Objective**

$$E(w) = R(w) + \lambda\, w^T w$$

$\lambda > 0$ controls regularization strength.

**Ridge Regression Solution**

$$w^\star = (\Phi^T \Phi + \lambda I_F)^{-1} \Phi^T y$$

**Multi-output:** $W^\star = (\Phi^T \Phi + \lambda I_F)^{-1} \Phi^T Y$

## 8 Kernel Methods

### Kernel Function

**Kernel**

$$k(x_i, x_j) = \phi(x_i)^T \phi(x_j)$$

Measures similarity without computing $\phi$ explicitly.

- **Polynomial:** $(x_i^T x_j + c)^d$
- **Gaussian:** $\exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$

### Kernel Matrix

$$K_{ij} = k(x_i, x_j) \in \mathbb{R}^{N \times N}$$

### Kernel Ridge Regression

**Kernel Ridge Solution**

$$\alpha^\star = (K + \lambda I_N)^{-1} y$$

**Prediction**

$$\hat{y} = y^T (K + \lambda I)^{-1}\, k(X, x)$$

where $k(X, x) = \big(k(x_1, x), \ldots, k(x_N, x)\big)^T$

**Multi-output:**

$$W^\star = \Phi^T (K + \lambda I)^{-1} Y, \quad \hat{y} = Y^T (K + \lambda I)^{-1} k(X, x)$$

**Kernel selection**

Choose kernel by **cross-validation**. No general best choice.